Product Requirement Document

Project Title: HireFlow – Decision Engine for Recruitment Workflows

Domain: Domain 1 – AI Systems & Agentic Workflow Automation

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1.0 Project Overview

1.1 Problem Statement

Mid-sized HR teams managing high-volume hiring (200+ applicants per role) are forced to spend over 60% of their time on manual screening, coordination, and decision reconciliation, significantly limiting their ability to scale hiring operations (Team BotFriday, 2025). This leads to delayed hiring cycles, missed top candidates, and increased workload pressure, especially when managing multiple roles simultaneously.

Despite the adoption of traditional Applicant Tracking Systems (ATS), these systems function primarily as passive data repositories, lacking the ability to **interpret hiring intent, adapt to role-specific requirements, or execute multi-step decision workflows**.

As a result:

- Time-to-hire is prolonged by fragmented coordination
- Candidate quality is inconsistent due to subjective evaluation
- Decision-making lacks transparency and traceability, especially under increasing regulatory pressure such as Personal Data Protection Act (PDPA) 2026

Existing recruitment systems assist in managing data, but **fail to actively make, adapt, and execute structured hiring decisions across the workflow.**

1.2 Target Domain

AI-Driven Recruitment Decision Systems and Agentic Workflow Automation.

1.3 Proposed Solution Summary

HireFlow is an AI-powered recruitment decision engine that actively manages and executes hiring workflows through a stateful reasoning system built on Z.AI GLM. It processes unstructured inputs such as CVs and job descriptions, orchestrates multi-step hiring stages, and reconciles AI-driven insights with human evaluations to ensure consistent, explainable, and high-fidelity hiring decisions.

Unlike traditional ATS platforms that act as passive systems of record, HireFlow functions as an active decision engine. At its core, the system models job requirements as **structured**

decision criteria such as technical depth, ambiguity tolerance and hiring priority, enabling context-aware and consistent evaluation across candidates.

This enables the system to:

- Perform **context-aware candidate evaluation** instead of keyword matching
- Maintain a **persistent reasoning state** across hiring stages
- Autonomously **execute workflow decisions** such as shortlist, review and schedule

This enables HireFlow to not only generate insights but also take actions within the workflow such as routing candidates, triggering interviews, and managing transitions without manual intervention.

2.0 Background & Business Objective

2.1 Background of the Problem

Modern recruitment workflows in mid-sized organizations remain heavily reliant on manual coordination and fragmented tools, despite the presence of Applicant Tracking Systems (ATS). Recruiters must still switch between resumes, emails, calendars, and interview feedback systems, resulting in inefficient and inconsistent hiring processes.

While early AI tools introduced automation, they largely function as "text-in, text-out" systems that generate outputs without maintaining decision context or workflow continuity.

With increasing regulatory pressure such as the Personal Data Protection Act (PDPA) 2026 and the "Candidate Bill of Rights," organizations are now required to ensure transparency, consistency, and traceability in hiring decisions—capabilities that traditional systems are not designed to support.

2.2 Importance of Solving This Issue

The inability to make consistent and scalable hiring decisions directly impacts organizational performance. Delayed hiring cycles result in lost access to top talent, while inconsistent evaluations lead to poor hiring outcomes and increased turnover costs.

As hiring volume increases, these inefficiencies compound, placing significant operational strain on HR teams and limiting their ability to scale effectively.

In the current labour market, organizations that fail to adopt structured, decision-driven hiring systems risk falling behind competitors in both speed and quality of talent acquisition.

2.2.1 Industry Economic & Operational Benchmarks

Recent industry data demonstrates the measurable impact of agentic AI systems in recruitment. The shift toward agentic AI systems is supported by measurable industry outcomes:

1. **Workload Reduction**:
   Organizations adopting agentic AI models achieve a **40–60% reduction** in manual back-office workloads by automating complex, multi-step workflows (Awad, Serry, & Vasquez, 2025).

2. **Accelerated Time-to-Hire**:
   Companies report **30–50% faster hiring cycles**, with AI-driven screening reducing time-to-shortlist significantly, enabling faster talent acquisition (Avetisyan, 2025).

3. **Cost Efficiency**:
   AI-powered recruitment reduces hiring costs by an average of 30% per hire, driven by reduced manual screening and 20–40% lower cost-per-hire through automated scheduling and outreach (Kumar, 2026).

4. **Strategic Resource Allocation**:
   44% of recruiters cite time savings as the main driver for AI adoption, while 58% highlight improved candidate sourcing as critical for maintaining a competitive hiring pipeline (Kumar, 2026).

2.2.2 HireFlow Value Realization (Product-Level Impact)

Building on these industry benchmarks, HireFlow is designed to deliver targeted performance improvements through its GLM-powered, stateful workflow engine:

1. **Hiring Speed**:
   HireFlow is designed to enable up to a 50% reduction in time-to-fill by automating candidate evaluation and interview coordination (Gillis & Simpson, 2025).

2. **Decision Consistency (Quality of Hire)**:
   AI-assisted screening achieves a 53% success rate in subsequent human interviews, compared to 29% in traditional processes, resulting in more reliable hiring outcomes (World Economic Forum, 2025).

3. **Recruiter Productivity**:
   The system is designed to autonomously handle up to 67% of initial candidate interactions autonomously, increasing recruiter capacity (Gillis & Simpson, 2025).

4. **Standardization & Transparency**:

AI-driven evaluation reduces human hiring bias by 37% (Batista, 2025), while providing structured reasoning traces to support compliance with PDPA 2026 and modern transparency standards.

2.3 Strategic Fit / Impact

HireFlow positions Z.AI GLM-5.1 as a **stateful decision engine** rather than a conversational tool, enabling structured and continuous decision-making across recruitment workflows.

While the AI recruitment market continues to grow, most solutions remain limited to automation or isolated AI features. HireFlow targets this gap by introducing **end-to-end decision orchestration**, where the system not only generates insights but also executes workflow actions.

The solution aligns with Malaysia's MyDIGITAL initiative and SDG 8.2 by improving labour productivity through intelligent automation.

As a result, HireFlow enables scalable, consistent, and transparent hiring operations beyond the capabilities of traditional ATS platforms.

3.0 Product Purpose

3.1. Main Goal of the System

To deploy an AI-powered decision engine that automates and executes early-stage hiring workflows, reducing manual coordination while improving decision consistency and hiring efficiency.

3.2. Intended Users (Target Audience)

1. **Talent Acquisition (TA) Managers**
   TA Managers overseeing multiple hiring pipelines can shift from manual coordination to strategic oversight, using the system to monitor pipeline health and resolve decision conflicts efficiently.

2. **Technical Interviewers / Hiring Managers**
   Technical interviewers and hiring managers can focus entirely on evaluating candidates rather than managing administrative tasks. The system provides structured evaluation insights derived from interview transcripts and AI reasoning, highlighting technical strengths and potential gaps. This reduces interviewer fatigue and ensures more consistent and objective evaluations across different interviewers.

3. **Candidates (The Talent)**
   Candidates receive automated updates and scheduling support from the system. They can schedule, reschedule, and track their application status in real time without delays. This always-available experience reduces uncertainty and significantly lowers candidate drop-off rates.

4.0 System Functionalities

4.1 System Description

The system operates as a stateful, agentic AI-driven recruitment orchestration engine that transforms unstructured inputs such as resumes, job descriptions, and interview notes into structured decision data. Using Z.AI GLM, it evaluates candidate-role alignment and executes multi-step workflow actions such as routing, review, and scheduling. The system maintains workflow state across hiring stages, enabling consistent, traceable, and explainable decision execution.

4.2 Key Functionalities

1. **Intelligent Resume Understanding**
   Extracts structured candidate profiles from unstructured resumes, including skills, experience, education, and relevant role indicators.

2. **Job Intent Modelling**
   Transforms job descriptions into structured hiring criteria, including seniority level, skill priorities, role expectations, and decision constraints, to support context-aware candidate evaluation.

3. **Candidate Matching Engine**
   Uses GLM reasoning to evaluate candidate-role alignment and generate structured match outputs, including fit scores, mismatch analysis, and recommendation signals.

4. **Workflow Orchestration Engine**
   Dynamically routes candidates through hiring stages using rule-guided decision branching whether it is shortlisted, reviewed or rejected based on model outputs and workflow state.

5. **Interview Scheduling Automation**
   Integrates with calendar and notification APIs to automatically schedule interviews, resolve slot availability, and confirm meeting logistic.

6. **Hybrid Interview Evaluation System**
   Includes a reconciliation layer that combines AI-generated insights with human evaluations to support more consistent, transparent, and balanced decision outcomes.

7. **Hiring Decision Engine**
   Generates structured hiring recommendations with supporting reasoning, confidence scores, and conflict flags for human review.

8. **Lifecycle & Onboarding Automation**
   Triggers onboarding workflows and manages post-hire lifecycle stages.

4.3 AI Model & Prompt Design

4.3.1 Model Selection

Z.AI GLM is selected as the core reasoning engine due to its ability to:

- perform structured multi-step reasoning
- generate schema-constrained JSON outputs
- support workflow-aware decision tasks
- maintain contextual consistency across hiring stages

4.3.2 Prompting Strategy

The system adopts a multi-step prompting strategy that decomposes recruitment workflows into structured, sequential reasoning tasks. Rather than handling the workflow in a single prompt, the system separates reasoning into three stages:

- **Extraction:** Converts unstructured CVs into structured candidate profiles
- **Reasoning:** Evaluates candidate-job alignment and determines workflow decisions
- **Structuring:** Transforms interview inputs into standardized evaluation outputs

This strategy is used because recruitment is a multi-stage decision problem that benefits from intermediate representations such as candidate profiles and structured job criteria. These representations improve consistency, traceability, and reduce hallucination risk across workflow stages. By enforcing structured JSON outputs at each stage, the system makes GLM responses more consistent, machine-readable, and directly usable by downstream components.

4.3.3 Context & Input Handling

The system is designed to handle large and unstructured inputs efficiently through a combination of **chunking, filtering, and caching strategies**:

- **Maximum Input Size:** Supports bounded input processing through controlled token limits, with large documents segmented before reasoning.
- **Chunking:** Large CVs and transcripts are split into smaller segments and processed sequentially
- **Filtering:** Irrelevant sections such as repetitive content and non-skill-related text are removed before processing
- **Caching:** Frequently reused data such as Job Intent representations are cached to reduce token usage

If inputs exceed system limits, they are either segmented into processable chunks or rejected with validation feedback when the structure is incomplete or unusable. This ensures efficient processing while maintaining response quality and cost control.

4.3.4 Fallback & Failure Behaviour

The system implements a **multi-layer failure handling mechanism** to ensure robustness and reliability:

- **Validation Layer:** All GLM outputs are validated against predefined JSON schemas to detect malformed or incomplete responses
- **Retry Mechanism:** Automatically retries up to a defined threshold if outputs are invalid or inconsistent
- **Hallucination Mitigation:** Outputs that are low-confidence, incomplete, or inconsistent with workflow rules are flagged for review
- **Human Escalation:** Persistent failures are routed to a manual HR review queue for intervention
- **Graceful Degradation:** If GLM is unavailable, the system maintains workflow continuity by pausing decision transitions rather than breaking the workflow

This design reduces the risk of workflow disruption caused by model failure while maintaining transparency and reliability in decision support.

5.0 User Stories & Use Cases

5.1 User Stories

- **As a TA manager handling multiple open roles with over 200 applicants each,** I want the system to automatically evaluate and recommend shortlisted candidates based on structured job intent, so that I can reduce manual screening time from several hours per day to under two hours while maintaining consistent hiring quality.
- **As a hiring manager** reviewing shortlisted candidates across multiple interview rounds, I want to see structured and explainable candidate comparisons based on AI-derived insights and interview feedback, so that I can make more consistent hiring decisions with less reliance on subjective judgment.
- **As a candidate,** I want real-time updates and flexible scheduling without delays, so that I can navigate the hiring process transparently and without uncertainty.

5.2 Use Cases

1. **Resume Processing Flow**
   User uploads resume → system parses and chunks input → GLM extracts structured candidate profile → profile validated → workflow instance created → candidate enters initial evaluation stage

2. **Candidate Matching & Routing Flow**
   System retrieves structured job criteria → GLM evaluates candidate-role alignment → generates match score and reasoning output → decision engine routes candidate to shortlist, review, or reject stage → workflow state is updated

3. **Interview Scheduling Flow (Agentic Behaviour)**
   System detects shortlisted candidate → checks interviewer availability via calendar API → resolves scheduling conflicts → proposes optimal time slots → confirms booking → sends automated notifications

4. **Interview Evaluation Flow (Human-AI Hybrid)**

Interviewer submits feedback → GLM structures and analyses the input → system reconciles AI-derived signals with human evaluation → generates a standardized interview assessment → candidate state is updated

5. **Hiring Decision & Onboarding Flow**
   System aggregates evaluation signals → GLM generates a structured hiring recommendation with reasoning → HR confirms the decision → downstream onboarding workflow may be triggered after approval

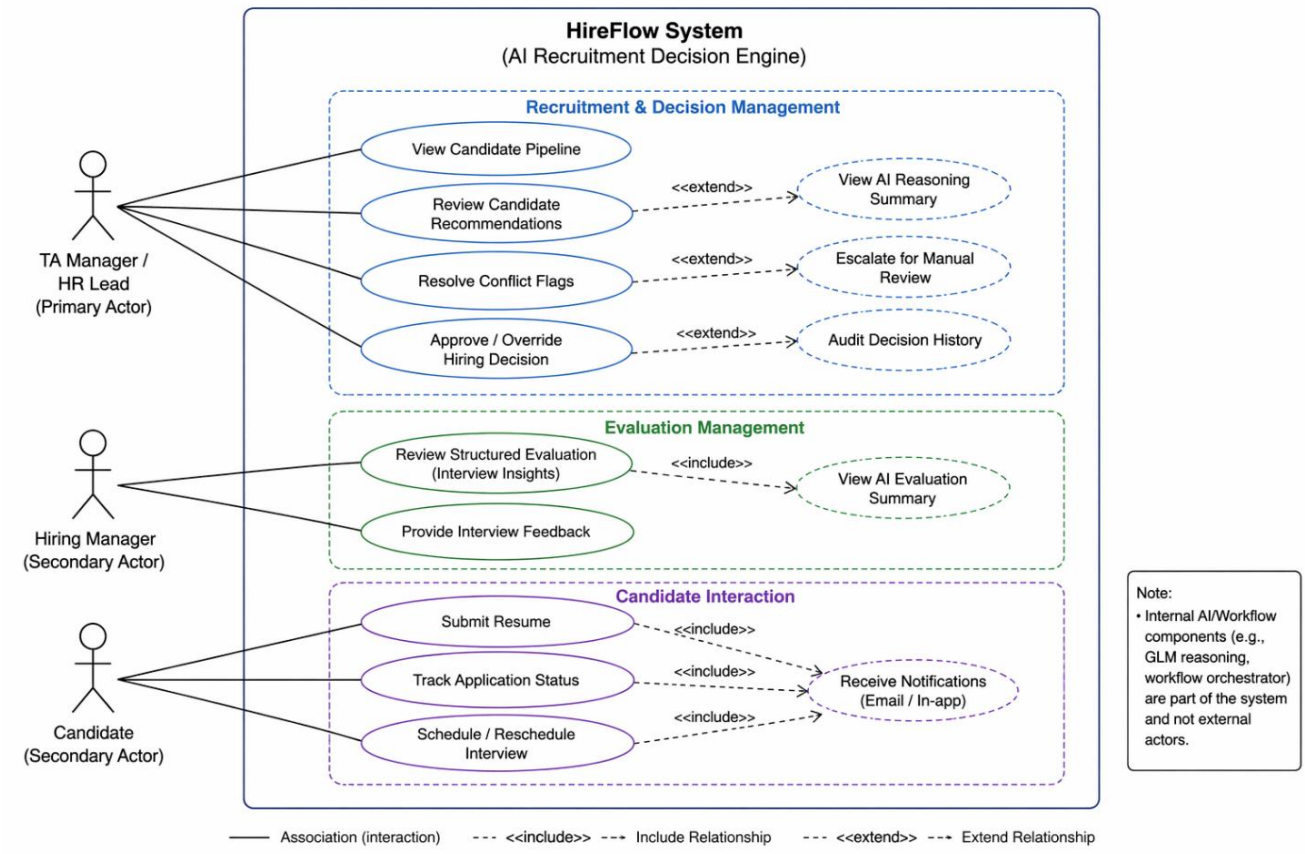
5.3 Core Product Loop

HireFlow improves over time through a continuous decision-feedback loop:

- More candidates processed →
- More structured evaluation data generated →
- Better alignment between structured job criteria and candidate signals →
- Improved decision consistency and recommendation quality →
- Increased recruiter trust and system reliance →

This creates a self-reinforcing cycle of better hiring quality and operational efficiency.

5.3 Use Case Diagram



6.0 Features Included (Scope)

HireFlow is designed as a stateful, AI-driven recruitment decision system with the following capabilities:

- Stateful workflow orchestration with multi-step reasoning
- Resume parsing and structured candidate profile extraction
- Job requirement modeling through structured decision criteria
- Context-aware candidate matching using semantic reasoning
- Decision routing across hiring stages (shortlist, review, reject)
- Interview scheduling automation with constraint-aware planning
- Hybrid evaluation combining AI insights with human feedback
- Structured hiring recommendations with explainable reasoning
- Dynamic onboarding workflow generation based on hiring outcome

7.0 Features Not Included (Scope Control)

The following features are intentionally excluded to maintain focus on the core hiring decision workflow:

- Payroll and compensation management systems
- Employee performance tracking and HR lifecycle analytics
- Full enterprise ATS integration (e.g., legacy system migration)
- Multi-company or marketplace-based recruitment platforms
- Real-time conversational chat systems with candidates

8.0 Assumptions & Constraints

1. **Technical Constraints**
   - System depends on external APIs such as email and calendar integration, which may introduce latency variability
   - GLM inference latency assumed stable at 3–5 seconds per reasoning call
   - Workflow execution is constrained to structured JSON-based communication between components
   - Free-form multi-agent communication is not implemented in the MVP
   - The system supports only text-based or extractable PDF resumes (no image-based OCR in MVP)

2. **Performance Constraints**
   - Multi-candidate batch processing is supported but prioritizes sequential workflow consistency for decision accuracy
   - Complex reasoning chains are capped (approximately 5–8 agent steps per candidate) to prevent latency issues

3. **User Input Constraints**
   - The system assumes that resumes and job descriptions are provided in text or machine-readable PDF formats

- Poorly structured or incomplete inputs may reduce extraction and matching accuracy
- Final hiring decisions require human approval in the MVP to ensure accountability and compliance

9.0 Risks & Questions Throughout Development

1. **Model Hallucination in Hiring Decision**

   Risk: The model may generate incorrect or unsupported outputs during candidate evaluation.

   Mitigation:

   - Structured JSON schema validation
   - Rule-based consistency checks
   - Controlled retry and refinement loops

2. **Multi-step Agent Reasoning Drift**

   Risk: Model outputs may deviate from job requirements across multiple workflow steps.

   Mitigation:

   - State locking mechanisms
   - Reinforcement of constraints at each workflow stage
   - Validation against structured job criteria

3. **Workflow State Corruption**

   Risk: Incorrect state updates may lead to inconsistent or invalid workflow transitions.

   Mitigation:

   - Persistent state storage with versioned snapshots
   - Rollback capability for state recovery

4. **API Failure (Email/Calendar Integration)**

   Risk: Failures in third-party APIs may disrupt scheduling and communication workflows.

   Mitigation:

   - Retry mechanisms with backoff strategy

- Asynchronous fallback queues for delayed execution

5. **High LLM Cost at Scale**

   Risk: Frequent reasoning calls may increase operational cost as usage grows.

   Mitigation:

   - Caching and reuse of intermediate outputs
   - Prompt compression techniques
   - Selective activation of reasoning for high-impact decisions only

6. **AI vs Human Decision Conflict**

   Risk: Discrepancies may occur between AI recommendations and human judgment.

   Mitigation:

   - Hybrid evaluation framework combining AI and human inputs
   - Explainability layer to surface reasoning and confidence levels
   - Final decision authority retained by human reviewers